

Integrated Barangay System — Source Code & Documentation

Gawad BIS (ASP.NET Core 8) + **BHC System** (PHP / MySQL)

Barangay Balong Bato, San Juan City

Audience: Developers, IT implementers, thesis reviewers, and maintainers.

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1. System overview

The integrated platform separates **barangay-wide records** from **clinical operations** while linking them where it matters:

Concern	System	Database
Resident master data, permits, barangay medicine inventory	Gawad BIS	MongoDB (gawad_db)
Patient registry, queues, triage, consultation, dispensing, clinical documents	BHC	MySQL (brgy_health_db)

Integration goals:

- One-time resident enrollment in Gawad → prefill BHC patient registration
- Shared staff identity (username) with optional SSO
- Read-only medicine catalog + stock from Gawad for BHC prescription picker
- Audit trail on both sides for accountability

Repository layout (typical):

bhc_system/	← BHC PHP app (this repository)
barangay-integrated-system/	
gawad-bis/	← junction or copy of Gawad project
project-gawad-main/	← Gawad BIS .NET solution (Project.Gawad.sln)

BHC lives at `c:\PUP\htdocs\bhc_system`. Gawad BIS is a separate .NET solution, often referenced via a workspace junction.

2. Technology stack

BHC System

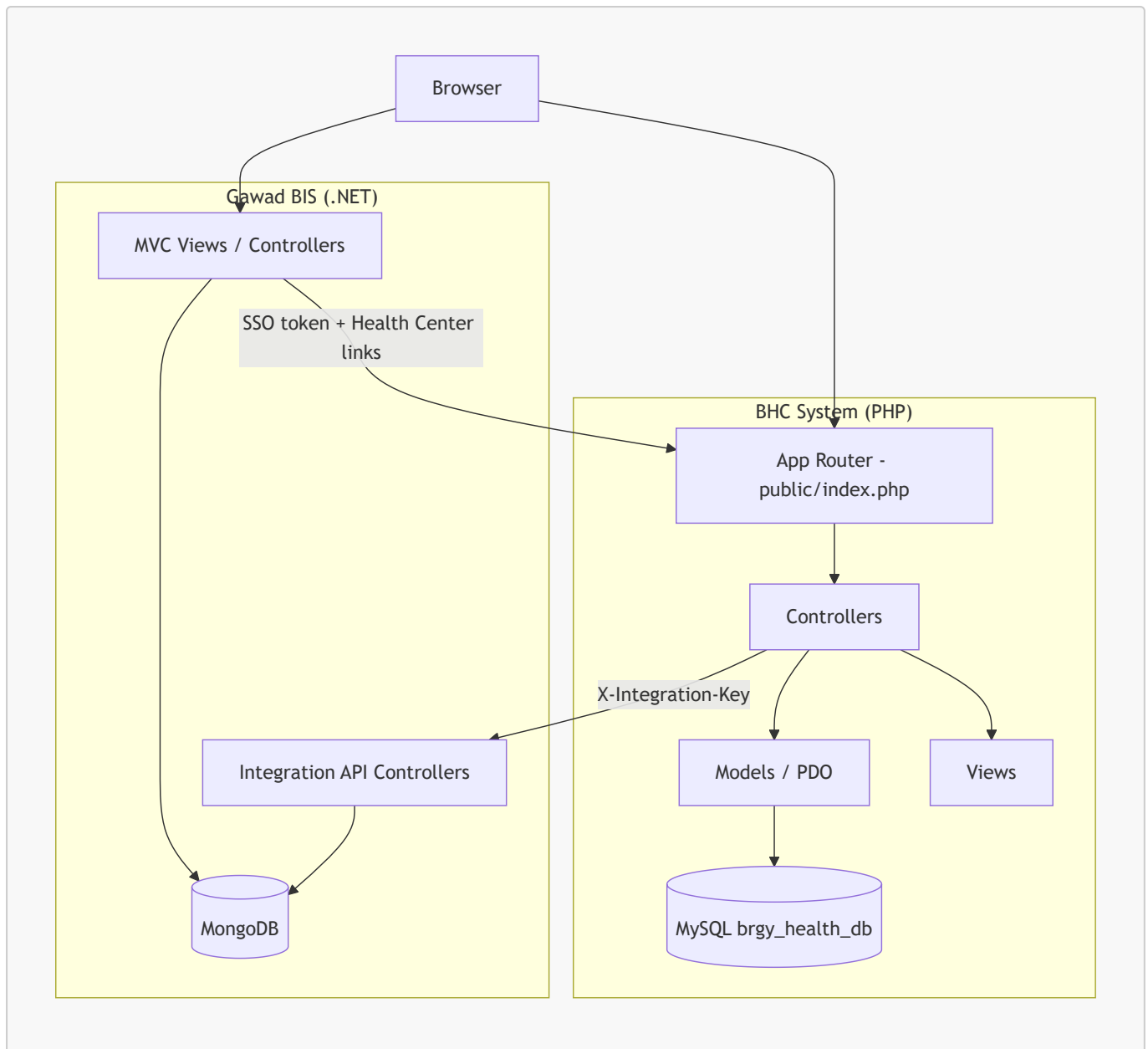
Layer	Technology
Language	PHP 8.1+
Architecture	Custom MVC (no framework)
Database	MySQL 8 / MariaDB (<code>utf8mb4</code>)
Web server	Apache (XAMPP) or InfinityFree
Session	PHP <code>\$_SESSION</code> via <code>core/Auth.php</code>
Front end	Server-rendered PHP views, vanilla JavaScript

Gawad BIS

Layer	Technology
Framework	ASP.NET Core 8 MVC (<code>net8.0</code>)
Database	MongoDB
Auth	ASP.NET Identity (cookies)
UI	AdminLTE, Bootstrap Icons
Dev URL	<code>http://localhost:5003</code>

3. Architecture

High-level diagram



Request flow (BHC)

1. Apache routes requests to `public/index.php` (`.htaccess` rewrite).
2. `core/App.php` matches method + path against registered routes.
3. Controller method runs (`requireAuth`, `requireRole`, business logic).
4. Model layer uses PDO (`config/database.php`).
5. View rendered via `Controller::view()`.

Design principles

- **BHC** owns clinical workflow and queue state.
- **Gawad** owns resident master data and medicine inventory.
- Integration is **HTTP + shared secrets**, not a shared database.
- BHC `ensureSchema()` in `config/database.php` migrates schema on connect (additive only).

4. BHC source code structure

```

bhc_system/
├── public/
│   ├── index.php          # Entry point + route table
│   ├── .htaccess          # URL rewriting
│   └── assets/            # CSS, JS, images, workflow diagrams
├── config/
│   ├── app.php            # Loads app.local.php or app.infinityfree.php
│   ├── app.local.php      # base_path, base_url (XAMPP)
│   ├── database.php       # PDO + ensureSchema()
│   └── gawad_integration.php + gawad_integration.local.php
├── core/
│   ├── App.php            # Router
│   ├── Controller.php     # Base controller, auth guards, redirects
│   ├── Auth.php           # Session login state
│   ├── helpers.php        # app_url(), app_route(), h()
│   ├── GawadIntegration.php
│   ├── GawadSso.php
│   ├── GawadStaffSync.php
│   ├── ReportMonth.php
│   └── ReportPeriod.php
├── controllers/           # 14 controllers (see table below)
├── models/                # PDO data access
├── views/                 # PHP templates + layout partials
├── scripts/               # CLI utilities
├── docs/                  # This documentation package
├── deploy/infinityfree/   # Production build output
└── database_schema.sql    # Canonical MySQL schema

```

Controllers

Controller	Responsibility
HomeController	Dashboards: / , /staff , /admin
AuthController	Login, logout, GET /auth/gawad (SSO)
PatientController	CRUD, search, Gawad import, link-gawad, archive
QueueController	Stations, queues, tickets, display, triage enqueue
CoordinatorController	Multi-station board /coordinator
ClinicalController	Triage/consultation/dispense at desk, documents, receipts
DoctorController	Doctor portal /doctor , patient charts, documents
AppointmentController	Clinic appointments
MedicineCatalogController	Local medicine name list (admin)
ReportController	Daily, monthly, clinical, appointments + CSV
AuditController	Activity log

Controller	Responsibility
<code>UserController</code>	Staff accounts, Gawad sync
<code>AccountController</code>	Password, document name (doctors)

Key models

Model	Table(s)
<code>Patient</code>	<code>patients</code> — search, validate, Gawad link
<code>QueueTicket</code>	<code>queue_tickets</code> — daily per-station queue
<code>PatientVisit</code>	<code>patient_visits</code> — one visit episode per day
<code>TriageRecord</code>	<code>triage_records</code>
<code>ConsultationRecord</code>	<code>consultation_records</code>
<code>MedicineDispensing</code>	<code>medicine_dispensings</code>
<code>MedicineCatalog</code>	<code>medicine_catalog</code> + Gawad picker overlay
<code>ClinicalDocument</code>	<code>clinical_documents</code>
<code>User</code>	<code>users</code>
<code>AuditLog</code>	<code>audit_logs</code>

Route registration

All routes are defined in `public/index.php` (lines 68–175). Examples:

Method	Path	Handler
GET	<code>/patients/create</code>	<code>PatientController::create</code>
GET	<code>/patients/search</code>	<code>PatientController::search</code> (JSON)
GET	<code>/auth/gawad</code>	<code>AuthController::gawadSso</code>
POST	<code>/users/sync-gawad</code>	<code>UserController::syncFromGawad</code>
GET	<code>/doctor/queue-snapshot</code>	<code>DoctorController::queueSnapshot</code> (JSON poll)
POST	<code>/queue/{id}/triage/{ticketId}</code>	<code>QueueController::storeTriage</code>
POST	<code>/queue/{id}/consultation/{ticketId}</code>	<code>ClinicalController::storeConsultation</code>
POST	<code>/queue/{id}/dispense/{ticketId}</code>	<code>ClinicalController::storeDispensing</code>

URL helpers

`core/helpers.php` provides subfolder-safe URLs:

- `app_url('/patients')` → `http://localhost/bhc_system/public/patients`
- `app_route('/patients')` → path for form actions
- `app_base_path()` → `/bhc_system/public`

`Controller::redirect()` prefixes `app_base_path()` automatically.

5. Gawad BIS source code structure

```
project-gawad-main/
├── Project.Gawad.sln
├── src/
│   ├── Project.Gawad.Client/           # MVC entry, Views, Integration
│   └── controllers
│       ├── Project.Gawad.Application/  # Services, BhcSsoLinkService, providers
│       ├── Project.Gawad.Core/         # Interfaces
│       ├── Project.Gawad.Domain/       # Entities, DTOs, enums
│       ├── Project.Gawad.Infrastructure/
│       └── Project.Gawad.Data.MongoDb/  # Repositories
```

Key Gawad files (integration)

File	Purpose
<code>Application/Options/BhcIntegrationOption.cs</code>	Config model
<code>Client/appsettings.json</code>	<code>BhcIntegration</code> section
<code>Client/Controllers/IntegrationResidentsController.cs</code>	Resident export API
<code>Client/Controllers/IntegrationMedicinesController.cs</code>	Medicine export API
<code>Client/Controllers/IntegrationUsersController.cs</code>	Staff export API
<code>Application/Services/BhcSsoLinkService.cs</code>	SSO token + BHC deep links
<code>Views/Residents/Profile.cshtml</code>	Register at Health Center button
<code>Views/Shared/PartialViews/_SideBarPartialView.cshtml</code>	Health Center sidebar links

Gawad roles (`RoleType`)

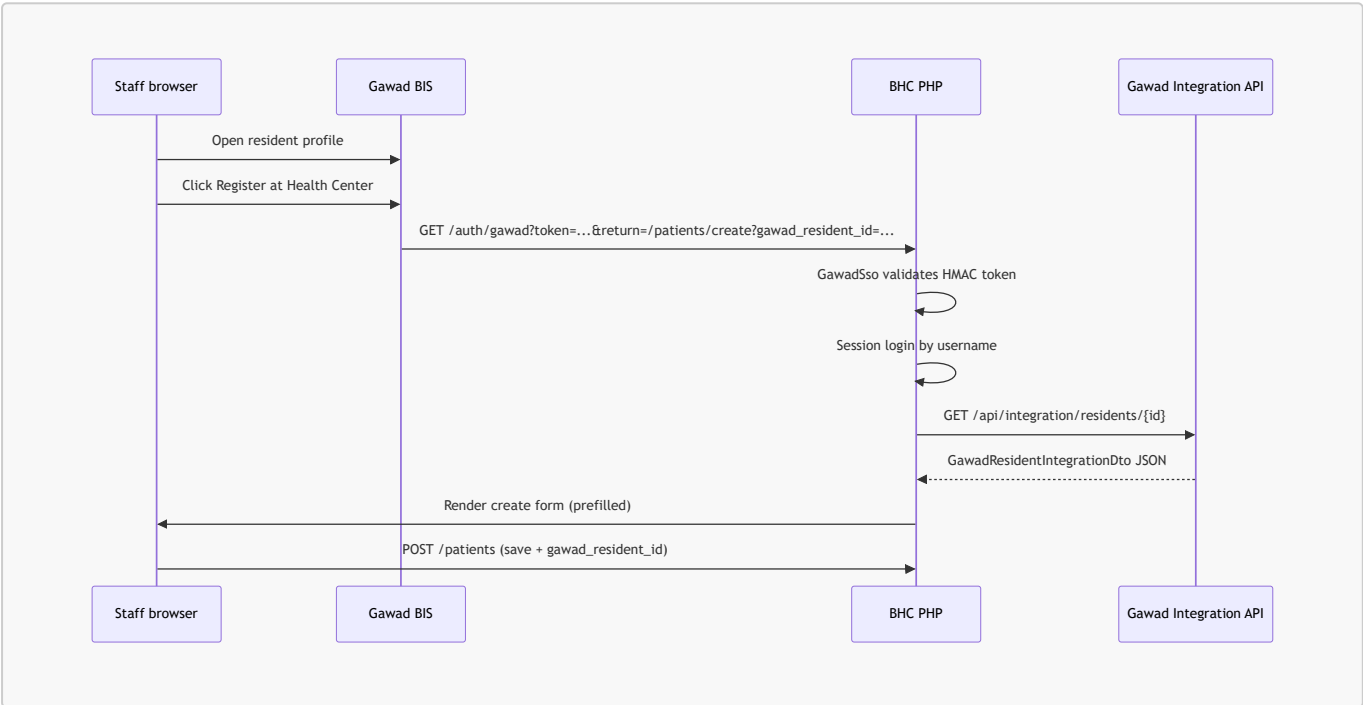
Enum	Display name
<code>Administrator</code>	Administrator
<code>BarangaySecretary</code>	Barangay Secretary
<code>Kagawad</code>	Kagawad
<code>Staff</code>	Health Worker / Staff

6. Integration design

Phase map

Phase	Feature	Direction
1	Deep links to BHC	Gawad → BHC (URLs)
2	Resident prefill	Gawad API → BHC <code>PatientController::create</code>
3	SSO	Gawad token → BHC <code>AuthController::gawadSso</code>
4	Medicine catalog sync	Gawad API → BHC <code>MedicineCatalog::pickerMeta</code>
—	Staff user sync	Gawad API → BHC <code>GawadStaffSync</code>

Resident registration sequence



Medicine sync behavior

- **Source of truth:** Gawad `Medicines` collection / stock batches.
- **BHC consumption:** `GawadIntegration::fetchMedicines()` → `MedicineCatalog::pickerList()`.
- **Fallback:** Local `medicine_catalog` table if API fails or sync disabled.
- **Dispense:** Recorded in BHC `medicine_dispensings`; does **not** POST back to reduce Gawad stock (by design — manual stock update in Gawad).

Staff sync behavior

- `GawadStaffSync::sync()` pulls `GET /api/integration/users`.
- Creates BHC users that do not exist (matched by `username`).
- Maps Gawad `Administrator` → BHC `admin`; others → `staff`.
- Does not create `doctor` roles or deactivate removed Gawad users automatically.

7. API reference

Authentication

All Gawad integration endpoints require:

```
X-Integration-Key: {shared secret}
```

Must match `BhcIntegration:IntegrationApiKey` (Gawad) and `integration_api_key` (BHC).

Gawad → BHC (outbound from Gawad)

SSO login URL pattern:

```
{BhcBaseUrl}/auth/gawad?token={signed}&return={urlencoded path}
```

Example return path: `/patients/create?gawad_resident_id=507f1f77bcf86cd799439011`

BHC → Gawad (inbound API calls)

Base URL: `{api_base_url}` (default `http://localhost:5003`)

GET `/api/integration/residents/{id}`

- **Gate:** `ResidentSyncEnabled`
- **ID format:** 24-char hex MongoDB ObjectId
- **Response fields:** `Id, FirstName, MiddleName, LastName, Suffix, Sex, Birthdate, Address, Barangay, CivilStatus, IsBarangayResident`, etc.
- **BHC consumer:** `core/GawadIntegration.php::fetchResident()`

GET `/api/integration/medicines`

- **Gate:** `Enabled + MedicineSyncEnabled`
- **Response:** Array of `{ Id, Name, UnitOfMeasure, CurrentStock, MinimumStockLevel, IsLowStock, IsOutOfStock }`
- **BHC consumer:** `GawadIntegration::fetchMedicines() → MedicineCatalog::pickerMeta()`

GET `/api/integration/users`

- **Gate:** `Enabled`
- **Response:** Array of `{ UserName, FullName, GawadRole, GawadRoleType }`
- **BHC consumer:** `GawadIntegration::fetchStaffUsers() → GawadStaffSync`

BHC JSON endpoints (internal)

Endpoint	Auth	Purpose
GET /patients/search?q=&birthdate=&contact=	Logged in	Duplicate patient autocomplete
GET /doctor/queue-snapshot	Doctor	Queue poll for auto-refresh
GET /patients/{id}/appointment-today	Logged in	Registration desk banner
GET /patients/{id}/queue-status	Logged in	Active ticket check

8. Single sign-on (SSO)

Token format

```
{base64url(payload)}.{base64url(hmac_sha256(payload, sso_secret))}
```

Payload JSON:

```
{
  "u": "username",
  "exp": 1717200000,
  "n": "random-nonce",
  "r": "optional-role-hint"
}
```

Gawad issuer

`BhcSsoLinkService` (Gawad) builds tokens when `SsoEnabled` and `SsoSecret` are set.

BHC validator

`core/GawadSso.php`:

1. Split token, verify HMAC with `sso_secret`.
2. Check `exp` against `sso_token_lifetime_seconds`.
3. Load BHC user by `u` (username).
4. `Auth::login()` and redirect to `return` path.

Failure modes

Condition	Result
Invalid/expired token	Login form with error
Username not in BHC	Login form — run staff sync
SSO disabled	Manual BHC login only

9. Database schemas

BHC (MySQL) — `database_schema.sql`

Core tables:

Table	Purpose
<code>patients</code>	Registry; <code>bhc_id</code> , <code>gawad_resident_id</code> , <code>registration_source</code> , residency fields, <code>archived_at</code>
<code>stations</code>	RG, TR, CN, PH (seeded)
<code>queue_tickets</code>	<code>status</code> : waiting serving done skipped
<code>patient_visits</code>	Daily visit episodes
<code>triage_records</code>	Vitals per visit
<code>consultation_records</code>	Diagnosis, <code>doctor_id</code>
<code>medicine_dispensings</code>	Line items per consultation
<code>medicine_catalog</code>	Local fallback names
<code>clinical_documents</code>	Receipts, certificates, referrals
<code>patient_appointments</code>	Follow-up scheduling
<code>users</code>	<code>role</code> : admin staff doctor
<code>audit_logs</code>	Structured activity trail

Runtime migrations in `config/database.php::ensureSchema()` add columns such as `gawad_resident_id`, `document_display_name`, `assigned_doctor_id`, etc., without dropping data.

Gawad (MongoDB)

Collections include (representative):

- Residents / persons
- Medicines, stock batches, transactions
- ASP.NET Identity users and roles
- Barangay operational modules (clearances, permits, etc.)

Gawad does not store BHC queue or clinical data.

10. Configuration

BHC — `config/gawad_integration.php`

Key	Description
<code>enabled</code>	Master switch for API calls

Key	Description
<code>api_base_url</code>	Gawad base URL
<code>integration_api_key</code>	Must match Gawad
<code>sso_enabled</code>	Accept <code>/auth/gawad</code> tokens
<code>sso_secret</code>	HMAC shared secret
<code>sso_token_lifetime_seconds</code>	Token TTL (default 300)
<code>medicine_sync_enabled</code>	Use Gawad medicines in picker

Local overrides: copy `gawad_integration.local.example.php` → `gawad_integration.local.php`.

BHC — `config/app.local.php`

```
'base_path' => '/bhc_system/public',  
'base_url'  => 'http://localhost/bhc_system/public',
```

For patient QR on phones, use the server **LAN IP** instead of `localhost`.

BHC — `config/database.php`

```
'host' => 'localhost',  
'dbname' => 'brgy_health_db',  
'user' => 'root',  
'password' => '',
```

Gawad — `appsettings.json` → `BhcIntegration`

```
{  
  "Enabled": true,  
  "BaseUrl": "http://localhost/bhc_system/public",  
  "OpenInNewTab": true,  
  "ResidentSyncEnabled": true,  
  "MedicineSyncEnabled": true,  
  "IntegrationApiKey": "balong-bato-dev-integration-key",  
  "SsoEnabled": true,  
  "SsoSecret": "balong-bato-dev-sso-secret",  
  "SsoTokenLifetimeSeconds": 300  
}
```

Both systems must use matching keys and secrets.

11. Local development setup

Prerequisites

- XAMPP (Apache + MySQL + PHP 8.1+)
- .NET 8 SDK
- MongoDB (local or Docker)
- Git

BHC

1. Place project in `htdocs/bhc_system/`.
2. Create MySQL database `brgy_health_db` (optional: import `database_schema.sql`).
3. Configure `config/app.local.php` and `config/database.php`.
4. Copy `config/gawad_integration.local.example.php` → `config/gawad_integration.local.php`.
5. Open `http://localhost/bhc_system/public/login`.
6. Default admin: `admin` / `admin123` (auto-seeded if no users).

Gawad BIS

1. Open `Project.Gawad.sln` in Visual Studio or VS Code.
2. Set `Project.Gawad.Client` as startup project.
3. Configure MongoDB connection string in `appsettings.json` / `appsettings.Development.json`.
4. Set `BhcIntegration` section (see above).
5. Run: `dotnet run --project src/Project.Gawad.Client` (or F5).
6. Browse `http://localhost:5003`.

Verify integration

```
# Medicines API (replace key)
curl -H "X-Integration-Key: balong-bato-dev-integration-key"
http://localhost:5003/api/integration/medicines

# BHC patient search (requires session cookie after login)
curl "http://localhost/bhc_system/public/patients/search?q=Juan"
```

Staff sync (CLI)

```
cd c:\PUP\htdocs\bhc_system
php scripts/sync_gawad_staff_users.php --password=TempPass123 --dry-run
php scripts/sync_gawad_staff_users.php --password=TempPass123
```

12. Deployment

BHC on InfinityFree

See [DEPLOY_INFINITYFREE.md](#):

1. Configure `config/database.infinityfree.php`.
2. Run `php scripts/build-infinityfree-deploy.php`.
3. Upload `deploy/infinityfree/package/` contents to `htdocs/`.
4. Production URL example: `https://bhcs.free.nf`.

Update Gawad `BhcIntegration:BaseUrl` to the production BHC URL if linking from a hosted Gawad instance.

LAN deployment (recommended for clinic)

- BHC on a fixed-LAN-IP XAMPP PC
- Gawad on same network or same server
- Waiting-area TVs use `/display/{stationId}` with LAN `base_url`

Rebuild deploy package

After code changes, always re-run the build script — the packaged copy may lag behind development.

13. Utility scripts

Script	Purpose
<code>scripts/create_staff_user.php</code>	CLI user creation
<code>scripts/sync_gawad_staff_users.php</code>	Pull Gawad users into BHC
<code>scripts/build-infinityfree-deploy.php</code>	Production package build
<code>scripts/seed_*.php</code>	Development data (if present)

14. Security notes

Topic	Current behavior	Production recommendation
Integration API key	Shared static key in config	Rotate keys; use env vars, not committed secrets
SSO secret	Shared HMAC secret	Strong random secret; HTTPS only
Gawad API	<code>[AllowAnonymous]</code> + API key header	Restrict by IP or mTLS in production
Default admin	<code>admin/admin123</code> on empty DB	Change immediately; deactivate default
Patient display	Public, no auth	Ticket numbers only; hide names on public TVs
SQL injection	PDO prepared statements	Keep pattern for new queries
CSRF	Forms use POST; no global CSRF token yet	Consider tokens for sensitive actions

Config directory is blocked by `config/.htaccess` on Apache.

15. Related documents

Document	File
Documentation index	README.md
Integrated user guide	INTEGRATED_USER_GUIDE.md
BHC user guide	USER_GUIDE.md
Workflow summary	WORKFLOW_SUMMARY.md
Workflow diagrams	WORKFLOW_DIAGRAM.md , workflow-diagram.mmd , ticket-lifecycle.mmd
InfinityFree deploy	DEPLOY_INFINITYFREE.md
MySQL schema	database_schema.sql (project root — not in this folder)

Appendix A — Default station IDs

ID	Code	Name
1	RG	Registration
2	TR	Triage / Vitals
3	CN	Consultation
4	PH	Pharmacy

Appendix B — BHC audit action types (sample)

`login`, `logout`, `login_failed`, `patient_create`, `patient_update`, `ticket_create`, `ticket_call`, `ticket_call_next`, `ticket_call_next_auto`, `ticket_complete`, `ticket_skip`, `ticket_recall`, `user_create`, `user_deactivate`, `user_activate`, `user_password_reset`, `gawad_staff_sync`, `gawad_link`

Appendix C — Patient registration sources

<code>registration_source</code>	Meaning
<code>gawad_bis</code>	Created via Gawad import link
<code>emergency_walk_in</code>	Staff emergency path with documented reason
<code>admin_direct</code>	Admin manual registration

Source Code & Documentation — Integrated Gawad BIS + BHC System. June 2026.